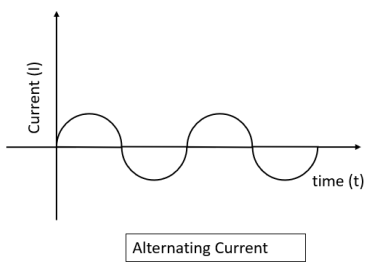
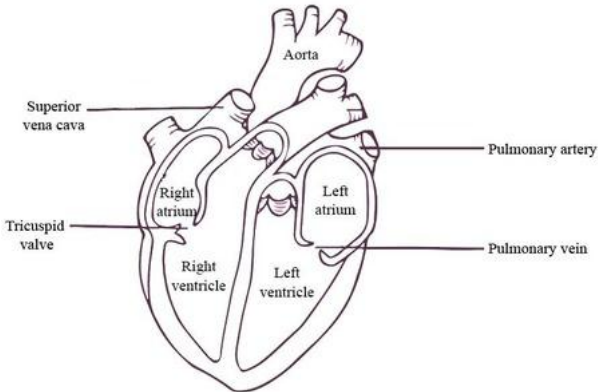


SEE 2082 (2026)
अनिवार्य विज्ञान तथा प्रविधि
उत्तरकुञ्जिका

यस कुञ्जिकामा भएको बाहेक अन्य सही वैकल्पिक तरिकाबाट समस्या समाधान गरेमा पनि अंक प्रदान गर्नुपर्ने छ ।
 उत्तरकुञ्जिकामा प्रत्येक चरणको प्राप्ताङ्क १ प्रदान गर्नु पर्नेछ तथा एकाइ छुट भएमा ०.५ अङ्क मात्र घटाउनुपर्ने छ ।

प्र. नं.	उत्तर	अंक
खण्ड 'क' (Section 'A')		
1 a.	(iv) Short wave radio receiver	1
b	(iv) Z	1
c	(iii) Fact and statement 1 are correct but statement 2 is incorrect	1
d	(i) Value of acceleration due to gravity decreases when we go higher from the surface of the earth or (ii) Value of acceleration due to gravity decreases when the height increases from the surface of the earth	1
e	(ii) 25N	1
f	(ii) p - critical angle and q- angle of reflection	1
g	(ii) 2p, 3s 3p, 4s	1
h	(iii) $3\text{Mg} + \text{N}_2 \xrightarrow{\Delta} \text{Mg}_3\text{N}_2$	1
i	(iv) Haematite	1
j	(iv) Statement and argument 1 are correct but argument 2 is wrong	1
खण्ड 'ख' (Section 'B')		
2.a	- To ensure that only independent variable affect dependent variables	1
b	- They both have mammary glands	1
c	- approx. 25N	1
d	- Falling of the object in the absence of any obstacle only under acceleration due to gravity	1
e	- Pascals Law	1
f		1
g	- Group II A or Group 2	1
h	- To separate magnetic and non magnetic impurities	1

2.i	H H H H-C-H-C-H OH OH OH Glycerol	1				
	खण्ड 'ग' (Section 'C')					
3	- LHS= P = J/s = kgm²s⁻²/s = kgm²s⁻³ RHS = W/t = mad/t = kgms⁻²xm/s = kgm²s⁻³ Thus, the equation is valid	1 1				
4	- Yes it is Possible, Leaf of Pteridophyta - feather like, Leaf of gymnosperm - needle like, angiosperm - broad leaf with parallel or reticulate venation	1 1				
5	<table><tr><td>Worker bee</td><td>Drone bee</td></tr><tr><td> 1. Collection of pollen grain/ nectar 2. 6 days to form larvae </td><td> 1. To participate in mating or fertilization 2. 7 days to form larvae </td></tr></table>	Worker bee	Drone bee	1. Collection of pollen grain/ nectar 2. 6 days to form larvae	1. To participate in mating or fertilization 2. 7 days to form larvae	1 for each
Worker bee	Drone bee					
1. Collection of pollen grain/ nectar 2. 6 days to form larvae	1. To participate in mating or fertilization 2. 7 days to form larvae					
6	- the medical or veterinary procedure of injecting semen into the vagina or uterus. - fast and selective reproduction or other benefits	1 mark for each				
7	Father Mother 44 + XY 44 + XX 22 X 22 Y 22 X 22 X 44 + XX 44 + XX 44 + XY 44 + XY Daughter Daughter Son Son	1 1				
8	- a) S - b) P	each 1 1				
9	- It is widely used as a traditional remedy to treat sexual dysfunction - Yarsagumba is used to fight fatigue and increase physical energy levels	each 1				
10	- Derivation of equation $F = 4G\ m_1x\ m_2/\ d^2$ - Explanation of equation	1 1				
11	3cm B A C F f 2cm v F₁ C₁ B' A' - Image formed is inverted	1 1				

12	a) - The recessional velocity of P is greater - because recessional velocity of P is greater than Q b) According to Hubble study universe is expanding	1 1
13	- Because the chemical reactivity increases with increment of their atomic radii	2
14	- 1) Temperature 2) Concentration	1 1
15	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$ METHANE - VectorStock VectorStock.com/2007035 - The IUPAC name is Methanol	1 1
16	- Use of vinegar in pickle - it doesn't hamper the health	1 1
खण्ड “घ” Section “D”		
17	- Positive impacts 1) It helps in study or to gain knowledge 2) Work become faster - Negative impacts 1) Brain becomes lazy 2) Eye strain or similar correct answers	2 2
18	 - if Pulmonary artery, Aorta and Right auricle or atrium is shown with neat and labelled diagram of heart just like shown above	4
19	- a) second generation b) phenotype ratio - 3:1, genotype ratio - 1:2:1 c) RR d) law of purity of gametes	1 1 1 1

20	a) substance C cools fastest because its specific heat capacity is least b) $Q = ms\Delta t$ or, $Q = 20 \times 900 \times 50$ or, $Q = 9,00,000 \text{ J}$	2 2
21	Given - Let, $N_2 = 50 N_1$ - using relation $V_2/V_1 = N_2/N_1$ - $V_2 = 11000 \text{ V}$ Ans - Step Transformer, used in Power houses	1 2 1
22	a) Dispersion of light b) A = Red B = violet c) A has more speed than B d) Change of speed B is more than A	1 1 1 1
23	a) Ammonia gas b) The volume of gas decreases due to presence of water vapour c) $2\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \longrightarrow \text{CaCl}_2 + 2\text{H}_2\text{O} + 2\text{NH}_3$	1 1 2